Sandeepa Fernando

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Personal Profile

Computational chemistry PhD student with hands on experience in molecular dynamic simulations, modeling biological systems and analyzing large scale simulation data. Able to independently identify and solve problems with a clear analytical approach, work effectively under time constrains situations and, manage multiple independent projects simultaneously.

Education

2024 – present: PhD in Chemistry

Wayne State University, Detroit, USA.
Advisor: Dr. Alice. R. Walker.
Key Research Interest: Molecular dynamic simulations, excited state QM/MM calculations, machine learning.

2019 – 2023: B. Sc(Honors) in Chemistry

University of Sri Jayawardenepura, Sri Lanka
Thesis: Analysis of calcium-dependent free energy landscape of calmodulin protein and a few of its mutants.
Minor: Genetics and Molecular Biology

Teaching Experience

Graduate Teaching Assistant (August 2024 - Present)

Wayne State University, Detroit

- Demonstrated experimental techniques and proper instrument use
- Supervised students during the lab, answered questions and troubleshooting any problems
- Graded students reports

Teaching Assistant(June 2023-April 2024)

Department of Chemistry, University of Sri Jayawardenepura, Sri Lanka.

- Assigned to undergraduate Physical Chemistry Laboratory and demonstrated laboratory practical
- Graded reports

Research Experience

Graduate Research Assistant (Spring/Summer 2026)

Wayne State University, Detroit

- Utilized computational methods and tools to understand the underlying mechanisms and photo physics of different fluorescent proteins to optimize and improve their functions

Projects/Research

Undergraduate research titles 'Analysis of calcium dependent free energy landscape of calmodulin protein and few of its mutants' | 2023-2024

University of Sri Jayawardenepura, Sri Lanka.

- Mapped the free energy landscape of calcium free and calcium bound calmodulin.
- Used mapped free energy landscape to Understand the folding pathway of calmodulin upon binding of calcium.
- Compared the free energy landscape of wild-type and mutated calmodulin to investigate the folding pathway changes with the mutations in calmodulin.

Awards

- Departmental citation for excellence in teaching services. (Wayne State University-2025)

Professional licenses and certifications

- SciPhD 2-Day Bootcamp: The Business of Science for scientists (May 2025)

Skills

Python, Linux (Ubuntu), Amber, Gromacs, VMD, Gaussian, Octave, MATLAB

References

Dr. Alice R. Walker
Principal Investigator
Department of Chemistry
Wayne State University
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